



torte: Reproducible Feature-Model Experiments à la Carte

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Implementing Configurable Software Systems

A Configurable Graph

```
class Node {  
    #ifdef LABELED  
        std::string label;  
    #endif  
    #ifdef COLORED  
        std::string color;  
    #endif  
};  
  
class Edge {  
    #ifdef DIRECTED  
        Node from, to;  
    #elif UNDIRECTED && HYPER  
        std::set<Node> nodes;  
    #endif  
};
```

Product Line Implementation

(here: C++ with C preprocessor)

Implementing Configurable Software Systems

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~~#define~~
LABELED
DIRECTED

Labeled Directed

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Configuration

Product Implementation

Implementing Configurable Software Systems

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Configuration

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Colored Undirected Hyper

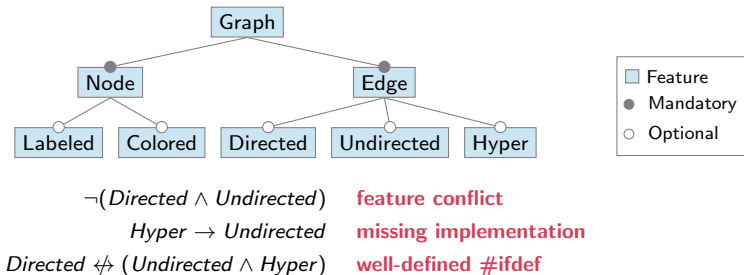
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Product Implementation

Modeling Features and their Dependencies

Feature Models

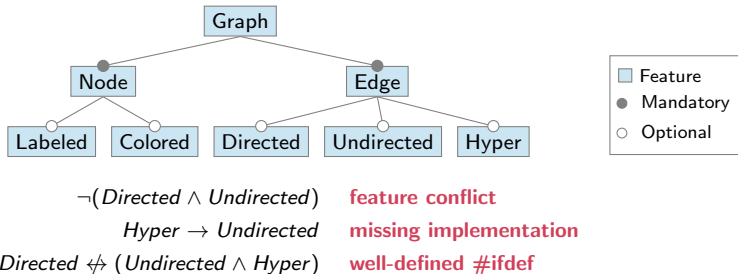
- tree models **features**
- cross-tree **constraints** model dependencies
- solver-based **analyses** can be used to understand the configuration space better



Modeling Features and their Dependencies

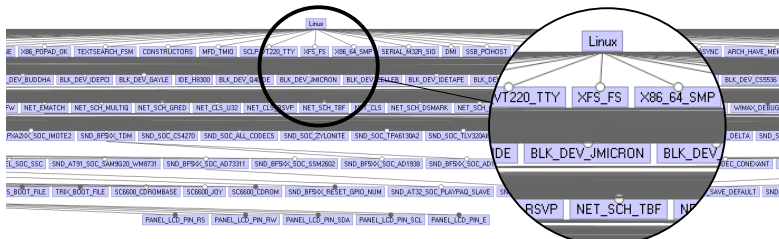
Feature Models

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The Linux Kernel

- ≈ 20000 features [2025]
- $> 10^{1600}$ products [2025]
- 114 dead features [2013]
- 151 reverse dependency bugs [2019]

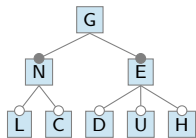


Feature-Model Analysis – Background

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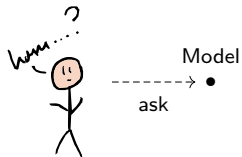


Feature Model



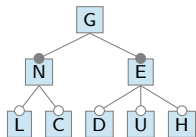
$\neg(D \wedge U)$
 $H \rightarrow U$
 $D \not\leftrightarrow (U \wedge H)$

Feature-Model Analysis – Background



----->
interactive

Feature Model

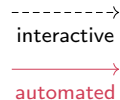
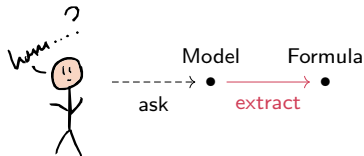


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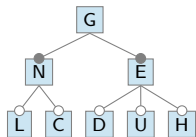
Core Feature F ?

#Configurations?

Feature-Model Analysis – Background



Feature Model



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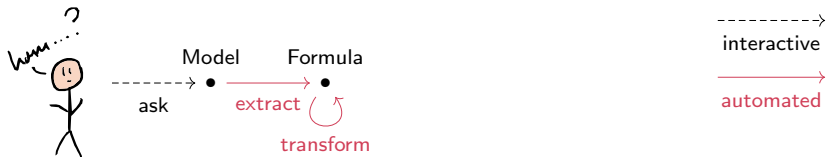
Formula

G
 $\wedge (N \leftrightarrow G) \wedge (E \leftrightarrow G)$
 $\wedge ((L \vee C) \rightarrow N)$
 $\wedge ((D \vee U \vee H) \rightarrow E)$
 $\wedge \neg(D \wedge U) \wedge (H \rightarrow U)$
 $\wedge (D \nleftrightarrow (U \wedge H))$

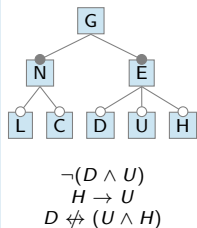
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Feature Model



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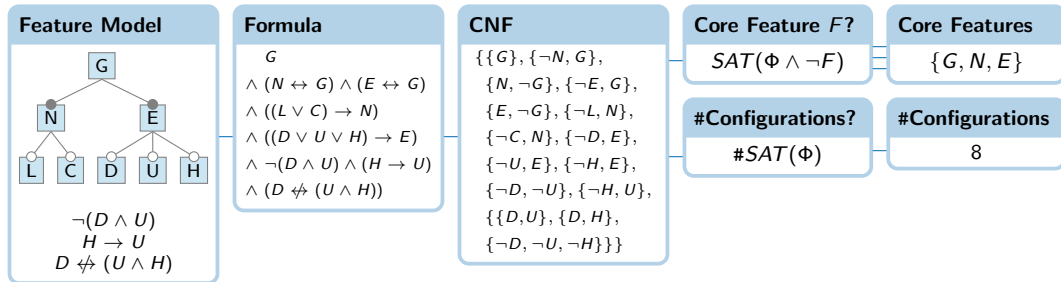
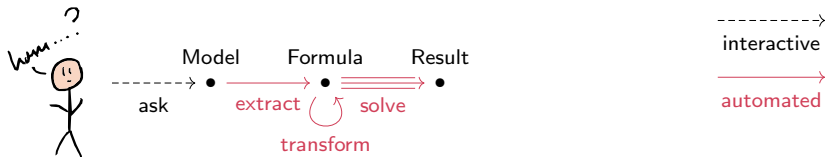
CNF

$\{\{G\}, \{\neg N, G\},$
 $\{N, \neg G\}, \{\neg E, G\},$
 $\{E, \neg G\}, \{\neg L, N\},$
 $\{\neg C, N\}, \{\neg D, E\},$
 $\{\neg U, E\}, \{\neg H, E\},$
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 $\{\{D, U\}, \{D, H\},$
 $\{\neg D, \neg U, \neg H\}\}$

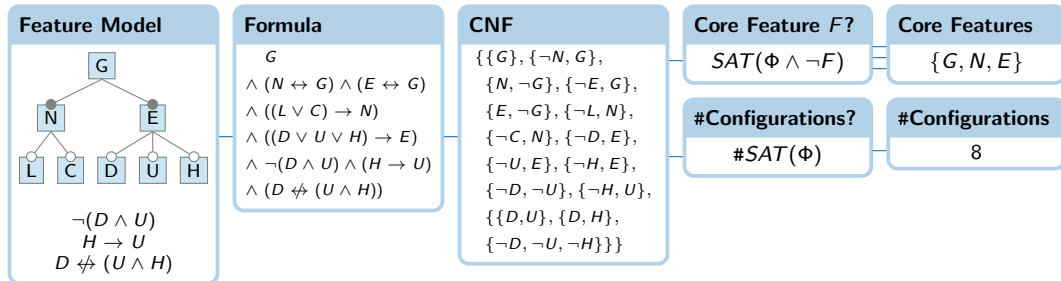
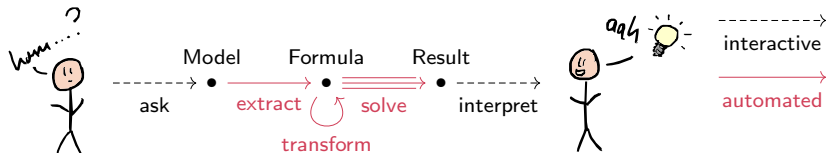
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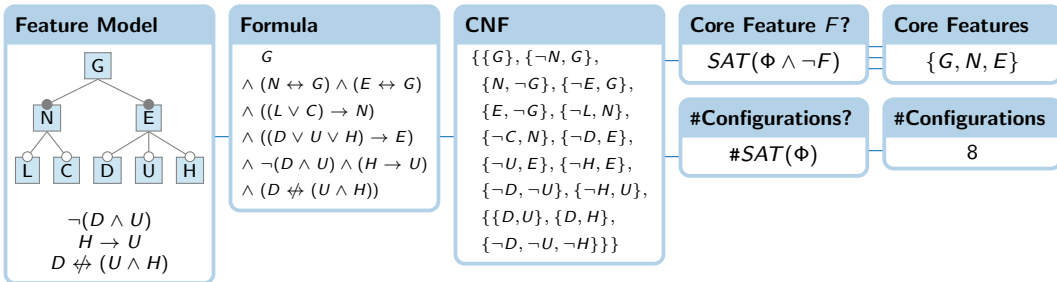
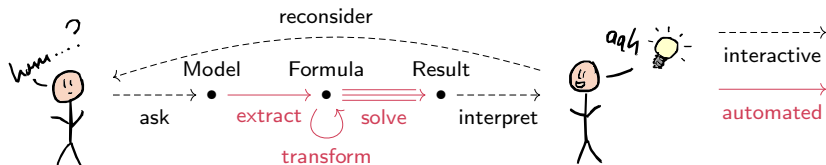
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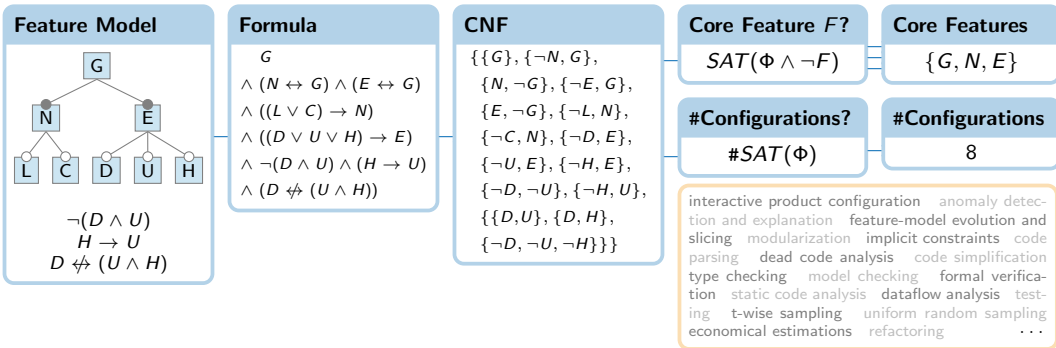
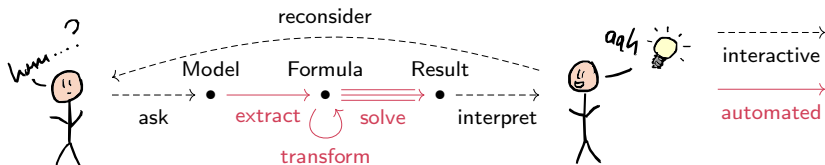
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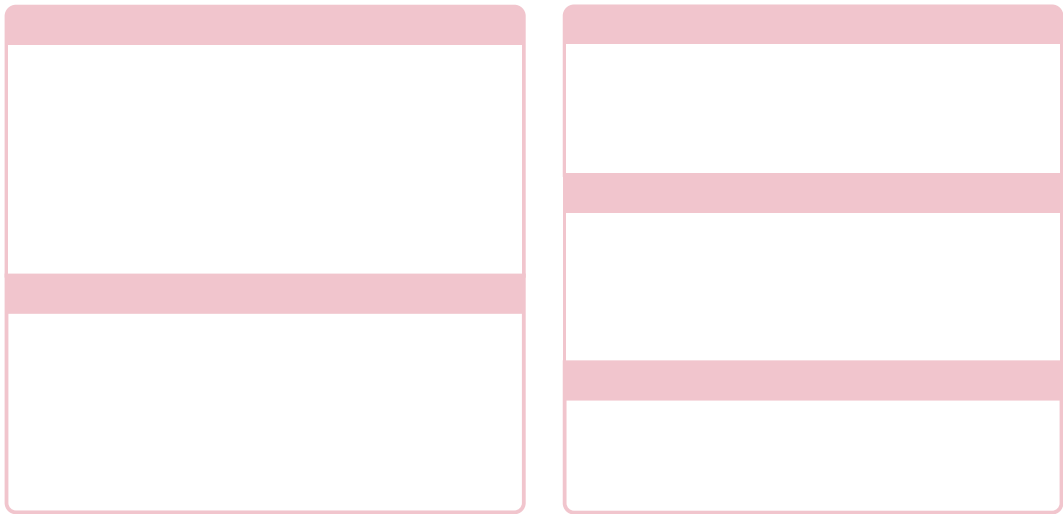
Feature-Model Analysis – Background



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Feature-Model Analysis – Open Challenges



Feature-Model Analysis – Open Challenges

History Extraction

BusyBox (20 years) axTLS (14 years) Buildroot (17 years) EmbToolkit (6 years) Freetz-NG (19 years) L4Re (9 years) toybox (19 years) uClibc (10 years) **Linux (24 years)** uClibc-ng (10 years) coreboot u-boot Crosstool-NG Soletta FreeWRT QEMU RIOT Unikraft Zephyr esp-idf sdk-nrf ...

KConfig **evolves/diverges** $\Rightarrow$ no consistent histories

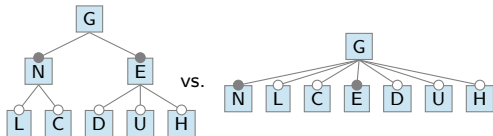
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Hierarchy Extraction



extracted models are **flat** $\Rightarrow$ loses developer intent

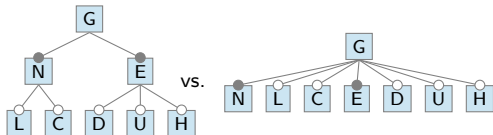
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Automation and Reproducibility

913bf31 · 10 years ago outdated dependencies

linked, for GNU/Linux 2.0.0, precompiled binaries

experiments are **rarely reproduced**

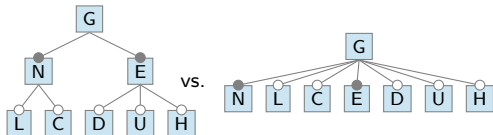
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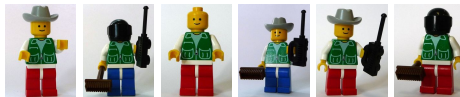
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Reusability and Interoperability



handcrafted experiments, limited **comparability**

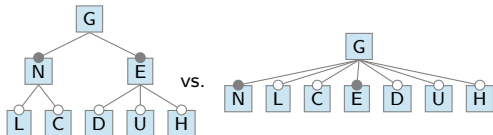
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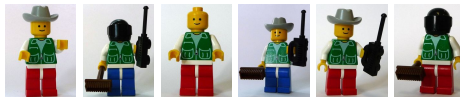
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Ease of Use

C(++) Scala Rust Java Bash Python ...

complex orchestration of heterogeneous tools



torte – An Experimentation Platform For Feature-Model Analysis

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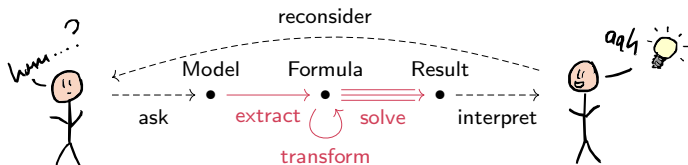
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Roll your own experiments!



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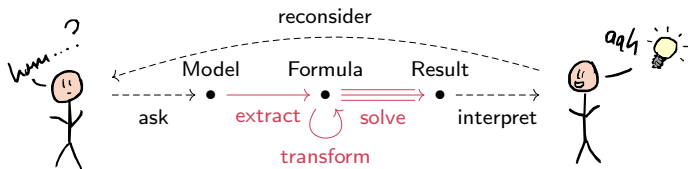
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Extraction

KCONFIGREADER
KCLAUSE
CONFIGFIX
Hierarchy extractor
Benchmark models

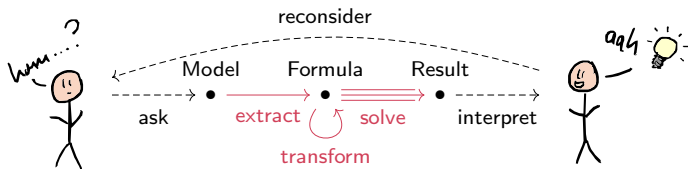
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torte – An Experimentation Platform For Feature-Model Analysis



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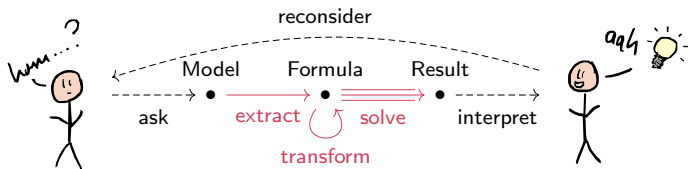
Roll your own experiments!



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Extraction	Transformation
KCONFIGREADER	FEATUREIDE
KCLAUSE	FEATJAR
CONFIGFIX	CLAUSY
Hierarchy extractor	Z3
Benchmark models	CADIBACK

torte – An Experimentation Platform For Feature-Model Analysis



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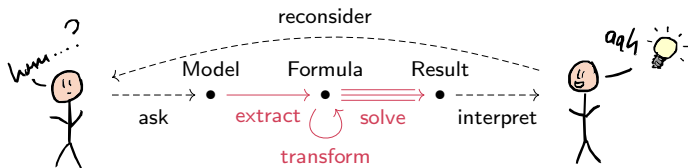
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Extraction	Transformation	Solving
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Experimentation Platform		
Docker containerization	Job parallelization	Stage architecture
Continuous integration	Profiling system	Experiment DSL
Reproduction packages	Resource limiting	Remote execution

torte – Short Demonstration

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How configurable is BusyBox?



torte – Short Demonstration

How configurable is BusyBox?



Self-Executing One-Liner

```
curl -sL https://elias-kuitter.de/torte/ | sh
```

Requirements Docker, Git, curl, and GNU tools

torte – Short Demonstration

```
#!/bin/bash
TORTE_REVISION=7a65a22
TIMEOUT=10
JOBS=4
[[ $TOOL != torte ]] && builtin source /dev/stdin <<<"$(curl -fsSL \
https://raw.githubusercontent.com/ekuiter/torte/7a65a22/torte.sh)"

experiment-systems() {
    add-busybox-kconfig-history --from 1_36_0 --to 1_36_1
}

experiment-stages() {
    clone-systems; read-statistics; extract-kconfig-models

    # Serializes the feature model of BusyBox as UVL
    transform-model-with-featjar \
    --transformer transform-model-to-uvl-with-featureide \
    --output-extension uvl
    transform-model-to-dimacs

    # Counts the number of valid configurations of BusyBox
    solve-sharp-sat --timeout "$TIMEOUT" --jobs "$JOBS"
}
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How configurable is BusyBox?



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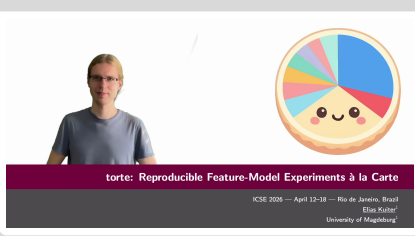


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Demo Video



Experiences with torte

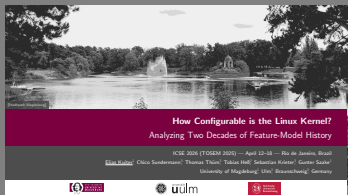
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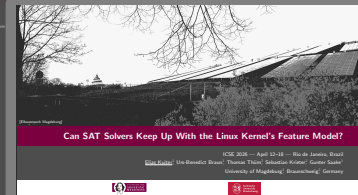
- 

Conclusion

Journal-First Paper



Research Paper



How Configurable Is the Linux Kernel?

Can SAT Solvers Keep Up With the Linux Kernel's Feature Model?

Demo Paper



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Image Credits

- [BusyBox logo](#), BusyBox developers [<https://commons.wikimedia.org/wiki/File:BusyBoxLogo.png>]
- [LEGO minifigures](#), Thomas Thüm [<https://github.com/TUBS-ISF/Course-on-Software-Product-Lines/tree/main/pics/lego>]

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